

- Description

This module manages a smart boiler water temperature control system, ensuring that:

1. The system continuously monitors the current water temperature (in degrees Celsius).
2. If the temperature falls below a predefined minimum threshold, the heating system is activated to increase the water temperature.
3. If the temperature exceeds a predefined maximum threshold, the heating system is deactivated to prevent overheating.
4. If the temperature is within the optimal range, no adjustment is made

This system helps maintain an optimal water temperature for safe and efficient operation.

- Specification

**module** SmartBoilerControl

**var** currentTemperature, minTemperature, maxTemperature: real

**var** heatingActive: boolean

**process** updateTemperature(newTemperature: real) returns (currentTemperature: real, heatingActive: boolean)

**functional scenario form:**

currentTemperature < minTemperature && heatingActive = true, output "ALERT: Water temperature too low (" + currentTemperature + "°C). Heating system activated.", Heating system activated.

||

currentTemperature > maxTemperature && heatingActive = false, output "ALERT: Water temperature too high (" + currentTemperature + "°C). Heating system deactivated.", Heating system deactivated.

||

currentTemperature <= maxTemperature && output "Temperature optimal (" + currentTemperature + "°C). No action needed.", No control action required.

**end process**

**end module**

- Java Code

```
package code;
public class c109_SmartBoiler {
    private double minTemperature;
    private double maxTemperature;
    private double currentTemperature;
    private boolean heatingActive;
    public c109_SmartBoiler(double minTemperature, double maxTemperature, double initialTemperature) {
        this.minTemperature = minTemperature;
        this.maxTemperature = maxTemperature;
        this.currentTemperature = initialTemperature;
    }
}
```

```

        this.heatingActive = false;
    }
    public void updateTemperature(double newTemperature) {
        currentTemperature = newTemperature;
        adjustHeatingSystem();
        System.out.printf("Current Water Temperature: %.2f°C, Heating:
%s%n",
            currentTemperature, heatingActive ? "ON" : "OFF");
        if (heatingActive) {
            System.out.println("Heating system activated to increase te
mperature.");
        } else {
            System.out.println("Temperature is within the optimal range
.");
        }
    }
    private void adjustHeatingSystem() {
        if (currentTemperature < minTemperature) {
            heatingActive = true;
        } else if (currentTemperature > maxTemperature) {
            heatingActive = false;
        }
    }
    public double getCurrentTemperature() {
        return currentTemperature;
    }
    public static void main(String[] args) {
        c109_SmartBoiler boiler = new c109_SmartBoiler(40.0, 80.0, 60.0);
        // testcase-VT:
        //     boiler.updateTemperature(35.0);
        //     boiler.updateTemperature(70.0);
        //     boiler.updateTemperature(85.0);

        // testcase-FT:
        //     boiler.updateTemperature(-1.7976931348623157E308);
        //     boiler.updateTemperature(0.0);
        //     boiler.updateTemperature(4.9E-324);

        // testcase-Z3:
        //     boiler.updateTemperature(35.0);
        //     boiler.updateTemperature(70.0);
        //     boiler.updateTemperature(85.0);

        // testcase-UVT:

```

```

        boiler.updateTemperature(35.0);
        boiler.updateTemperature(70.0);
        boiler.updateTemperature(85.0);
    }
}

```

- Coverage report

1. Test case number: 3

2. V-Method testing branch coverage: 100%

| Element                          | Coverage | Covered Branches | Missed Branches | Total Branches |
|----------------------------------|----------|------------------|-----------------|----------------|
| ▼ VibrationTestingTool           | 1.0 %    | 8                | 834             | 842            |
| ▼ src                            | 1.0 %    | 8                | 834             | 842            |
| ▼ code                           | 1.0 %    | 8                | 834             | 842            |
| ▼ c109_SmartBoiler.java          | 100.0 %  | 8                | 0               | 8              |
| ▼ c109_SmartBoiler               | 100.0 %  | 8                | 0               | 8              |
| ● updateTemperature(double)      | 100.0 %  | 4                | 0               | 4              |
| ■ adjustHeatingSystem()          | 100.0 %  | 4                | 0               | 4              |
| ● c109_SmartBoiler(double, doubl |          | 0                | 0               | 0              |
| ● main(String[])                 |          | 0                | 0               | 0              |

3. Vibration testing instruction coverage: 96.7%

| Element                          | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|----------------------------------|----------|----------------------|---------------------|--------------------|
| ▼ VibrationTestingTool           | 0.4 %    | 89                   | 21,917              | 22,006             |
| ▼ src                            | 0.4 %    | 89                   | 21,917              | 22,006             |
| ▼ code                           | 0.4 %    | 89                   | 21,917              | 22,006             |
| ▼ c109_SmartBoiler.java          | 96.7 %   | 89                   | 3                   | 92                 |
| ▼ c109_SmartBoiler               | 96.7 %   | 89                   | 3                   | 92                 |
| ● updateTemperature(double)      | 100.0 %  | 37                   | 0                   | 37                 |
| ■ adjustHeatingSystem()          | 100.0 %  | 20                   | 0                   | 20                 |
| ● c109_SmartBoiler(double, doubl | 100.0 %  | 15                   | 0                   | 15                 |
| ● main(String[])                 | 100.0 %  | 17                   | 0                   | 17                 |

4. Fuzz testing branch coverage: 37.5%

| Element                          | Coverage | Covered Branches | Missed Branches | Total Branches |
|----------------------------------|----------|------------------|-----------------|----------------|
| ▼ VibrationTestingTool           | 0.4 %    | 3                | 839             | 842            |
| ▼ src                            | 0.4 %    | 3                | 839             | 842            |
| ▼ code                           | 0.4 %    | 3                | 839             | 842            |
| ▼ c109_SmartBoiler.java          | 37.5 %   | 3                | 5               | 8              |
| ▼ c109_SmartBoiler               | 37.5 %   | 3                | 5               | 8              |
| ● updateTemperature(double)      | 50.0 %   | 2                | 2               | 4              |
| ■ adjustHeatingSystem()          | 25.0 %   | 1                | 3               | 4              |
| ● c109_SmartBoiler(double, doubl |          | 0                | 0               | 0              |
| ● main(String[])                 |          | 0                | 0               | 0              |

5. Fuzz testing instruction coverage: 82.6%

| Element                          | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|----------------------------------|----------|----------------------|---------------------|--------------------|
| ▼ VibrationTestingTool           | 0.3 %    | 76                   | 21,930              | 22,006             |
| ▼ src                            | 0.3 %    | 76                   | 21,930              | 22,006             |
| ▼ code                           | 0.3 %    | 76                   | 21,930              | 22,006             |
| ▼ c109_SmartBoiler.java          | 82.6 %   | 76                   | 16                  | 92                 |
| ▼ c109_SmartBoiler               | 82.6 %   | 76                   | 16                  | 92                 |
| ● updateTemperature(double)      | 89.2 %   | 33                   | 4                   | 37                 |
| ■ adjustHeatingSystem()          | 55.0 %   | 11                   | 9                   | 20                 |
| ● c109_SmartBoiler(double, doubl | 100.0 %  | 15                   | 0                   | 15                 |
| ● main(String[])                 | 100.0 %  | 17                   | 0                   | 17                 |

6. Z3 branch coverage: 100%

| Element                          | Coverage | Covered Branches | Missed Branches | Total Branches |
|----------------------------------|----------|------------------|-----------------|----------------|
| ▼ VibrationTestingTool           | 1.0 %    | 8                | 834             | 842            |
| ▼ src                            | 1.0 %    | 8                | 834             | 842            |
| ▼ code                           | 1.0 %    | 8                | 834             | 842            |
| ▼ c109_SmartBoiler.java          | 100.0 %  | 8                | 0               | 8              |
| ▼ c109_SmartBoiler               | 100.0 %  | 8                | 0               | 8              |
| ● updateTemperature(double)      | 100.0 %  | 4                | 0               | 4              |
| ■ adjustHeatingSystem()          | 100.0 %  | 4                | 0               | 4              |
| ● c109_SmartBoiler(double, doubl |          | 0                | 0               | 0              |
| \$ main(String[])                |          | 0                | 0               | 0              |

7. Z3 instruction coverage: 96.7%

| Element                          | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|----------------------------------|----------|----------------------|---------------------|--------------------|
| ▼ VibrationTestingTool           | 0.4 %    | 89                   | 21,917              | 22,006             |
| ▼ src                            | 0.4 %    | 89                   | 21,917              | 22,006             |
| ▼ code                           | 0.4 %    | 89                   | 21,917              | 22,006             |
| ▼ c109_SmartBoiler.java          | 96.7 %   | 89                   | 3                   | 92                 |
| ▼ c109_SmartBoiler               | 96.7 %   | 89                   | 3                   | 92                 |
| ● updateTemperature(double)      | 100.0 %  | 37                   | 0                   | 37                 |
| ■ adjustHeatingSystem()          | 100.0 %  | 20                   | 0                   | 20                 |
| ● c109_SmartBoiler(double, doubl | 100.0 %  | 15                   | 0                   | 15                 |
| \$ main(String[])                | 100.0 %  | 17                   | 0                   | 17                 |

8. Unimproved V-method branch coverage: 100%

| Element                          | Coverage | Covered Branches | Missed Branches | Total Branches |
|----------------------------------|----------|------------------|-----------------|----------------|
| ▼ VibrationTestingTool           | 1.0 %    | 8                | 834             | 842            |
| ▼ src                            | 1.0 %    | 8                | 834             | 842            |
| ▼ code                           | 1.0 %    | 8                | 834             | 842            |
| ▼ c109_SmartBoiler.java          | 100.0 %  | 8                | 0               | 8              |
| ▼ c109_SmartBoiler               | 100.0 %  | 8                | 0               | 8              |
| ● updateTemperature(double)      | 100.0 %  | 4                | 0               | 4              |
| ■ adjustHeatingSystem()          | 100.0 %  | 4                | 0               | 4              |
| ● c109_SmartBoiler(double, doubl |          | 0                | 0               | 0              |
| \$ main(String[])                |          | 0                | 0               | 0              |

9. Unimproved V-method instruction coverage: 96.7%

| Element                          | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|----------------------------------|----------|----------------------|---------------------|--------------------|
| ▼ VibrationTestingTool           | 0.4 %    | 89                   | 21,917              | 22,006             |
| ▼ src                            | 0.4 %    | 89                   | 21,917              | 22,006             |
| ▼ code                           | 0.4 %    | 89                   | 21,917              | 22,006             |
| ▼ c109_SmartBoiler.java          | 96.7 %   | 89                   | 3                   | 92                 |
| ▼ c109_SmartBoiler               | 96.7 %   | 89                   | 3                   | 92                 |
| ● updateTemperature(double)      | 100.0 %  | 37                   | 0                   | 37                 |
| ■ adjustHeatingSystem()          | 100.0 %  | 20                   | 0                   | 20                 |
| ● c109_SmartBoiler(double, doubl | 100.0 %  | 15                   | 0                   | 15                 |
| \$ main(String[])                | 100.0 %  | 17                   | 0                   | 17                 |